

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

MSE 4201
(Materials for Energy Conversion and Storage)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Growing worldwide demand for energy, and problems of scarcity and environmental impact associated with conventional sources are at the base of a very probable energy crisis in the next two or three decades. So, what needs to be done to solve the world's increasingly urgent energy problems? (12)
- (b) Explain the significance of bandgap energy on the solar cell performance. (08)
- (c) Which factors limit the theoretical conversion efficiency of solar cell? How can the conversion efficiency of solar cell be improved? (15)
2. (a) Classify different types of solar cells. How does a solar cell convert sunlight to electricity? (15)
- (b) "Despite the promising economic advantages, a-Si still has major roadblocks to overcome"- Justify the statement. (08)
- (c) Explain the working principle of Dye-sensitized solar cell with a suitable figure. (12)
3. (a) Among all thin-film solar cells which solar cell has the highest performance in the laboratory? Explain the construction of that solar cell with a suitable figure. (10)
- (b) Draw a tri-layer electrode structure of alkaline fuel cell and explain the functions of different layer. (10)
- (c) Explain the fundamental principles of Direct Methanol fuel cell. (10)
- (d) Write down the functions of Electron transport layer and Hole transport layer of perovskite solar cell. (05)
4. (a) Why alkaline electrolyte fuel cells must be free from CO₂, while molten carbonate fuel cells need CO₂ in air to work? Explain briefly. (10)
- (b) Explain the favorable characteristics of a solid oxide fuel cell. (07)
- (c) Describe the design features of the Rooppur nuclear power plant reactor. (08)
- (d) Discuss the fabrication of the future reactor fuels. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Describe the major components of an electrochemical cell with their functions using necessary figure. (10)
(b) Suggest a suitable battery for emergency power on nuclear submarines and describe the charging-discharging action during energy storage using proper sketches. (12)
(c) Explain the working mechanism with necessary figure of an energy storage battery which is considered one of the most promising candidates for stationary electric energy storage. (13)
6. (a) "Li-ion batteries are applicable for large sized loads such as hybrid electric vehicles, pure electric vehicles"-Justify the statement. (08)
(b) Which are currently the most popular electrolytes for Li-ion batteries among liquids, polymers, and high-temperature solid-state electrolytes? Justify your answer. (12)
(c) "Zinc-air batteries are metal-air batteries powered by oxidizing zinc with oxygen from the air"-explain with necessary figure. (15)
7. (a) Why does electric double layer capacitor (EDLC) show high capacitance? Describe the fundamental principles of EDLC with necessary figure. (15)
(b) Why pseudocapacitor is a kind of supercapacitor, which is different from double layer capacitor? Briefly explain. (09)
(c) Why are noncentrosymmetric materials important in piezoelectricity? How does piezoelectricity work? (11)
8. (a) Describe the effect of TiC and transition metal oxides on hydrogen sorption kinetics of simple metal hydride with necessary figures. (15)
(b) "The most promising hydrogen storage media is metal hybrids"-do you agree with the statement? Justify your answer. (08)
(c) Describe the hydrogen adsorption mechanism in carbonaceous materials. What are the key parameters for increasing H₂ storage capacity in CNTs? (12)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

MSE 4211
(Professional Ethics)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) What is Engineering Ethics? Explain the scope of ethics in professional workplace. (10)
(b) Using the Socratic account of professionalism, can engineering be qualified as a profession? If so then explain how? (13)
(c) Read the following example carefully and identify whether it is good work or ordinary positive engineering. "A group of General Electric engineers on their own time in the late 1930s developed the sealed beam headlight, which greatly reduced the number of accidents caused by night driving. There was considerable doubt as to whether the headlight could be developed, but the engineers persisted and finally achieved success." (12)
2. (a) Distinguish between aspirational ethics and preventive ethics. (10)
(b) Justify the statement that the engineer is a professional. (12)
(c) What is moral dilemma? Discuss with illustrations, the various situations by which moral dilemmas may occur among engineers. (13)
3. (a) What is technology transfer? How do Multinational corporations act as an agency of technology transfer? Explain briefly. (13)
(b) Explain how the engineer is inspired by the codes of ethics. (12)
(c) Briefly describe normative inquiry and factual inquiry. (10)
4. (a) What do you mean by sustainable development? How do you relate sustainable development to professional ethics? Explain briefly. (13)
(b) Write a short note on: (i) Hacking and (ii) Phishing. (10)
(c) Briefly discuss the categories of computer crimes. How can you save yourself from the computer crimes? (12)

Section B

(Answer ANY THREE questions from this section in answer script B)

5. (a) David Parkinson is a member of the Madison County Solid Waste Management Planning Committee (SWMPC). State law requires that one of the committee members be a solid waste expert, David's area of specialization. SWMPC is considering recommending a specific plot of land in a sparsely populated area of Madison County for a needed public landfill. However, next to this site is a large tract of land that a group of wealthy Madison County residents wish to purchase in order to develop a private golf course surrounded by luxurious homes. Although small, this group is highly organized and it has managed to gather support from other wealthy residents in the county, including several who wield considerable political power. (20)

Informally recognized as the Fairway Coalition, this influential group has bombarded the local media with expensive ads in its public campaign against the proposed landfill site, advocating instead a site that borders on one the least affluent areas of Madison City. The basic argument is that a landfill at the site SWMPC is considering will destroy one of Madison County's most beautiful areas. Although as many as 8000 of Madison City's 100,000 residents live within walking distance of the site proposed by the Fairway Coalition, they lack the political organization and financial means to mount significant opposition.

SWMPC is now meeting to discuss the respective merits of the two landfill sites. Members of the committee turn to David for his views on the controversy.

How can David solve this problem? Analyze from the point of view of utilitarianism and respect for persons.

- (b) Can utilitarianism provide a moral justification for engineers who work for tobacco companies, for example, in designing cigarette making machinery? (15)

In your answer take account of the following facts (and others you may be aware of). Bangladesh Cancer Society in association with Dhaka University, American Cancer Society and Cancer Research UK conducted the study on 10,000 families in all districts of the country in between July 2017 and April 2018. According to the study, tobacco usage causes 1,00,961 deaths every year in Bangladesh, which is more than the combined deaths caused by alcohol and drug abuse, car accidents, homicide, suicide, and acquired immunodeficiency syndrome (AIDS) and second-hand smoke causes 24,757 deaths. Besides, over 70 lakh people - age 30 and above - suffer from various types of diseases related to tobacco usage. Cigarette companies do much good by providing jobs, through taxes, and through philanthropy. Most new users of cigarettes in Bangladesh are teenagers (younger than eighteen years of age).

6. (a) How does dishonesty occur in professionals research work? (07)

- (b) In the past, engineering societies have generally portrayed participation by engineers in unions and collective bargaining in engineering as unprofessional and disloyal to employers. Critics reply that such generalized prohibitions reflect the excessive degree to which engineering is still dominated by corporations' interests. What options might be pursued, and would they still involve "collective coercive action"? (18)

"Managers at a mining and refinery operation have consistently kept wages below industry-wide levels. They have also sacrificed worker safety to save costs by not installing special structural reinforcements in the mines, and they have made no effort to control excessive pollution of the work environment. As a result, the operation has reaped larger-than-average profits. Management has been approached both by individuals and by representatives of employee groups about raising wages and taking the steps necessary to ensure worker safety, but to no avail. A nonviolent strike is called, and the metallurgical engineers support it for reasons of worker safety and public health."

- (c) How does conflict arise in a person when he acts as both engineer and manager? Explain briefly. (10)
7. (a) On January 28, 1986, the Space Shuttle Challenger was destroyed in a disastrous fire shortly after liftoff. All passengers aboard the vehicle were killed. A presidential commission was formed to investigate the cause of the accident and found that the O-ring seals had failed, and, furthermore, that the seals had been recognized as a potential hazard for several years prior to the disaster. Roger Boisjoly an engineer of Morton Thiokol, discovered about the O-ring malfunction. Then he presented the problem to higher authorities. Roger consulted colleague about the problem, then ran some test and, gave final presentation and recommendation not to launch. But Morton Thiokol overrule Roger's recommendation not to launch. After the accident, Boisjoly gave testimony before the Presidential Commission on the Space Shuttle Challenger accident. Boisjoly's testimony raises the issue of whistleblowing and the extent of the legitimacy of loyalty of an engineer to the organization in which he or she is employed. Morton Thiokol blamed Boisjoly and argued, he seeped confidential information. As a result, Boisjoly and other engineers who testified to commission were demoted.
- Had Boisjoly violated the obligation of loyalty to his firm? Was his whistleblowing justified? Defend your answer.
- (b) Define gift, bribe and moonlighting. How do these are the forms of potential conflicts of interest? (15)
8. (a) In which situations can whistleblowing be ethical? Describe briefly. (10)
- (b) How does copyright differ from patent? (07)
- (c) Henry is in a position to influence the selection of suppliers for the large volume of equipment that his firm purchases each year. At Christmas time, he usually receives small tokens from several salesman, ranging from inexpensive ballpoint pens to a bottle of liquor. This year, however, one salesman sends an expensive briefcase stamped with Henry's initials. Should Henry accept the gift? Should he take any further course of action? Justify your answer. (18)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

MSE 4231
(Biomaterials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Nitinol is a great choice for making vascular stents. Explain with diagram. (13)
(b) Why is corrosion on biomaterials a big concern? What kinds of precautions should be taken to prevent this problem? (12)
(c) Why 316L is more favorable for implant than 316 grade? (10)
2. (a) Compare the pros and cons of wrought and casting Ti-6Al-4V, both of which are used for making implants. (10)
(b) Identify the problems that may arise if pure magnesium is used for biomaterial device. What is the potential counter measures to solve the problem? (15)
(c) Which method is used for manufacturing ASTM-F75 Co-Cr alloy? Describe the method with diagram. (10)
3. (a) Why are shape memory alloys used in orthodontics? (10)
(b) How does Cr induce intergranular corrosion in stainless steel? What kind of prevention should be taken to remove intergranular corrosion? (13)
(c) What are the differences between as elastic materials and viscoelastic materials? Explain how the rate of loading affects the behaviors of viscoelastic materials. (12)
4. (a) Why is the fatigue a great concern for biomaterial? What types of prevention should be taken to avoid this problem? (13)
(b) Differentiate between alpha and alpha beta Ti alloy. (10)
(c) Ceramic material is a good choice for manufacturing dental implant. Specify the reasons. (12)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Define bioceramic materials. Write down the characteristics of bioceramic materials. (10)
(b) Differentiate between Bioinert and Bio-active Bio-ceramics. (12)
(c) Define bioactive glass. Explain the mechanism of bone regeneration of bioactive glass with figures and chemical reactions. (13)
6. (a) Briefly explain the requirements of an ideal scaffold for bone tissue engineering. (12)
(b) What are the properties that required for dental, ophthalmic, orthopedic, cardiovascular, and drug delivery applications of Polymeric Biomaterials? (12)
(c) Define pyrolytic carbon with necessary figures. Write down its properties and applications as biomaterials. (11)
7. (a) What are the factors that influence the degradation of bio-ceramic? (13)
(b) What is biodegradable biopolymer? Write down the mechanism of biodegradation of biopolymers. (12)
(c) Write down the formula for the ceramic commonly present in dental enamel. What are the advantages of synthetic HAp? (10)
8. (a) What is degree of swelling and why it is important? (07)
(b) What is smart hydrogel? Write down about p^H -responsive hydrogels. (10)
(c) Write down different sterilization process of biopolymers. (10)
(d) Briefly explain the limitations of hydrogels. (08)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

EEE 4227
(Semiconductor Materials and Devices)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) What is semiconductor? Describe the differences of conductor and insulator with it using energy levels. (10)
(b) Differentiate between intrinsic and extrinsic semiconductor. Using necessary diagrams, show the formation of *P* and *N* type semiconductors. (10)
(c) Describe the effect of application of reverse voltage on a p-n junction. (08)
(d) Define drift and diffusion current. (07)
2. (a) What is zener diode? Briefly explain the zener breakdown mechanism. (08)
(b) Define transition and diffusion capacitance with necessary diagrams. (07)
(c) Draw the output waveform of the following circuits shown in figure 2(c). (20)

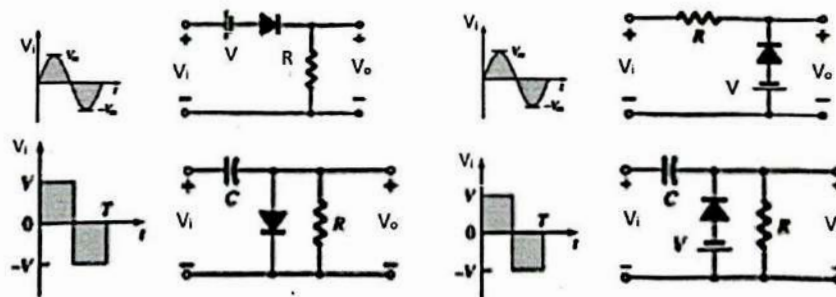


Figure 2(c)

3. (a) Describe the operation of full wave bridge rectifier with necessary diagram. (10)
(b) What is transistor? Prove that $I_c = 250 I_{CBO}$, where the symbols have their usual meanings. (10)
(c) Differentiate the followings for the fixed bias configuration shown in figure 3(c). (15)
(i) I_{BQ} and I_{CQ} (ii) V_{CEQ} (iii) V_B and V_C (iv) V_{BC} .

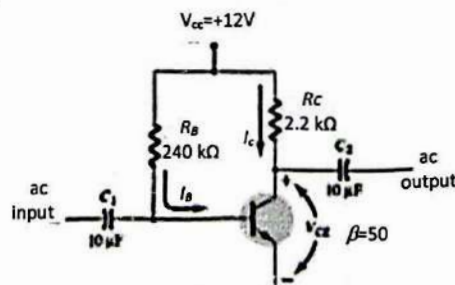


Figure 3(c)

4. (a) Write down the general technique to draw ac equivalent of transistor circuit and hence draw the ac equivalent circuit of common emitter configuration. (08)
- (b) What are the differences between *BJT* and *FET*? (06)
- (c) Explain the construction and operation of *n*-channel *JFET* using necessary diagrams. (15)
- (d) Sketch the transfer characteristics for a *p*-channel *JEFT* with $I_{DSS} = 4 \text{ mA}$ and $V_p = 3V$. (06)

Section B

(Answer ANY THREE questions from this section in answer script B)

5. (a) What is *MOSFET*? Why is the *MOSFET* called as voltage controlled device? (06)
- (b) Explain the construction and operation of *D-MOSFET* in both depletion and enhancement mode using necessary diagrams. (12)
- (c) Illustrate how the power dissipation is reduced in *VMOS*? Also mention its applications. (07)
- (d) Using the following 'ON Characteristics' data and an average threshold voltage $V_{GS(TH)} = 3V$, determine : (i) The '*k*' factor of the *MOSFET* $I_{D(on)} = 3 \text{ mA}$
(ii) The transfer characteristics. $V_{GS(on)} = 10 \text{ V}$ (10)
6. (a) Draw the circuit diagram of an *E-MOSFET* common source amplifier and explain its operation. (10)
- (b) Explain how does a *CMOS* act as digital switch. (06)
- (c) What are *MESFET* and Schottky barrier? Describe the characteristics of *MESFET*. (09)
- (d) Describe the mechanism behind the working of Light Emitting Diode (*LED*). Why is the *GaAs* material suitable for *LEDs* rather than *Si*, *Ge*? (10)
7. (a) Explain the operating principle of *PN* photodiodes. Also, distinguish between avalanche and *PiN* photodiodes. (12)
- (b) Discuss on the materials used to construct photodiodes and phototransistors. (05)
- (c) How does *LCD* work? Which materials are suitable for *LCD*? Explain in brief. (08)
- (d) An *InGaAs PiN* photodiode has the following parameters at a wavelength of 1300 nm: $I_D = 4 \text{ nA}$, $\eta = 0.90$, $R_L = 1000 \Omega$ and the leakage current is negligible. (10)
The incident power is 300 nW (-35 dBm) and receiver bandwidth is 20 MHz. Find (i) mean-square shot noise current, (ii) mean-square dark current, and (iii) mean-square thermal noise current.
8. (a) What is *SCR*? Why is Silicon used for *SCR* construction? (08)
- (b) Write down the construction features of *DIAC*. Also, draw the I-V characteristics of *DIAC*. (10)
- (c) Describe the construction and operation if *TRIAC*. (10)
- (d) Compare tunnel diode with normal *p-n* junction diode. (07)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

Hum 4227
(Project Management and Entrepreneurship)

Time: 3.0 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Define project management. Write down the steps in managing a project. (15)
(b) Suppose you are about to start a project. What will be your duties and responsibilities as a project manager? How will you set project objectives? (10)
(c) What is project charter? How will you define your project charter? (10)
2. (a) Suppose the Government of Bangladesh has offered you a project. The financial returns are not as lucrative as other corporate-backed projects available. What factors you should consider to accept or reject the project? Why you think your decision is wise? (10)
(b) You are a financial analyst for Dantor Ltd. You have two project proposals Project-A and Project-B. Each project has a cost of \$10,000 and cost of capital for each project is 12%. The projects expected cash flows are as follows: (25)

Year	Project-A	Project-B
1	6,500	3,500
2	3,000	3,500
3	3,000	3,500
4	1,000	3,500

Required:

- (i) Calculate projects' payback period and net present value.
 - (ii) Calculate Internal Rate of Return (IRR) of the projects.
 - (iii) Which project should be accepted and why?
3. (a) Describe the techniques that can be used to assess project progress. (10)
(b) The following table shows project activities in days- (15)

Activity	a	m	b	Predessors
a	5	7	12	—
b	8	8	8	—
c	2	6	10	a
d	12	14	19	a, b
e	6	6	12	c, d
f	3	12	18	b, e
g	6	8	10	f

Required:

- (i) Draw network diagram; and find out critical path and its completion time.
 - (ii) Find out the probability of completing the project in 55 days.
 - (iii) What is the completion time of the project corresponding to 95% probability?
- (c) What is WBS? What are the uses of Gantt chart in project management? (10)

4. (a) Define risk and uncertainty. What are the sources of project risks? Analyze. (10)
- (b) What might be the reasons for the termination of a project? (10)
- (c) How project risks can be addressed? How budgeting and project success are affected by intensity of project risks? (15)

Section B

(Answer ANY THREE questions from this section in answer script B)

5. (a) "Entrepreneurship or job"- what should engineers opt for? Why? (10)
- (b) Entrepreneurs are doers not thinkers. Argue and justify your stance. (10)
- (c) Are the successful entrepreneurs always born with some special entrepreneurial traits? Or you think some skills can be acquired through perseverance? Choose one and give your arguments in favor of your opinion. (15)
6. (a) How do you see entrepreneurship as a career option? What are your favorite aspects of being an entrepreneur? (10)
- (b) Define innovation from entrepreneurial aspect. What are the ways that an entrepreneur can undertake to be innovative while launching a new venture? (15)
- (c) Differentiate between business idea and business opportunity. What are the common sources of new business ideas? (10)
7. (a) 'Developing a formal business plan by investing a lot of money and time is just a wastage, a start-up investor should grab the earliest opportunity and launch the venture'-do you agree? Take a position and put arguments favoring your position. (10)
- (b) Briefly discuss the potentials and limitations of women entrepreneurship in Bangladesh. How the limitations can be resolved? (15)
- (c) How do entrepreneurs participate in economic growth initiatives? (10)
8. (a) What are the financing options usually available for funding a new business? (10)
- (b) What is venture capital? Discuss major financing limitations for a start-up. (15)
- (c) Evaluate penetration strategies and diversification strategies as the growth strategies of a new business venture. (10)